Fiscal Year:	FY 2014	**Task Last Updated:**	FY 08/27/2014
PI Name:	Ray, F. Andrew Ph.D.		
Project Title:	NEXGEN Approaches to Chromatid Painting		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Fort Collins	**State:**	CO
Zip Code:	80524	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Crew Health NNJ11ZSA002NA
Start Date:	09/01/2012	**End Date:**	04/30/2014
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date shows as 4/30/2014 per NSSC information (Ed. 3/26/14) NOTE: Extended to 03/31/2014 per NSSC information (Ed. 1/22/14) NOTE: Extended to 12/31/2013 per NSSC information (Ed. 11/12/13)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX12AM92G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	There is every reason to believe that large and small scale chromosomal intra-changes (inversions) are just as important as inter-changes between chromosomes, e.g., translocations, the latter of which are well known to cause cancer. And yet, inversions, especially small ones, have been historically difficult, if not impossible to detect. Recently, KromaTiD Inc., using SBIR funding from NASA, successfully demonstrated a new platform technology to detect small chromosomal inversions using directional oligonucleotide probes called chromatid painting. Here, we propose to develop the next generation of whole chromatid paints using a much faster, more cost effective approach. If successful, this will facilitate the rapid development of additional chromatid paints, with the goal of ultimately covering the entire genome. These next generation chromatid paints will be of even higher resolution than currently possible, so they will greatly expand potential biodosimetric applications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We are developing and refining a technology that allows retrospective assessment of previous DNA damage within an individual as measured by intrachromosal inversions. These mutations are highly stable in the somatic cells of an individual and can serve as biomarkers of previous double stranded DNA breaks long after the damage has occurred. Beyond monitoring DNA damage in astronaut's blood after space missions, the technology will be useful in professions where personnel may be exposed to radiation or clastogenic agents. The technology will also allow discovery of inversions that are associated with human diseases such as cancer and autism.
Task Progress:	This work was a collaborative effort between NASA Health Exploration Research Opportunity (HERO) programs and KromaTiD Inc., a small firm specifically focused on developing more sensitive cytogenetic assays to monitor prior exposure to ionizing radiation. The work described here is particularly relevant to the explicitly stated Space Radiation Risks and Gaps surrounding Radiation Carcinogenesis. In this context, NASA's interest lies in improving the detection of misrepaired, radiation-induced DNA damage and correlating it to the increased risk of cancer later in life. KromaTiD Inc. is engaged in developing additional directional Genomic Hybridization (dGH) chromatid paints to be used to implement a service branch capable of performing biodosimetry and discovery applications. Prior to this omnibus award, Kromatid Inc. was awarded a Small Business Innovative Research award from NASA. The overall goal of the SBIR was to develop a method to measure the production of intrachromosomal rearrangements called chromosome inversions and to apply the methodology as another means to assess past exposure to radiation. The project was successfully completed in July 2011. Before starting the SBIR, chromosome inversions were not easily detected by existing techniques and small inversions were virtually never detected. In general, inversions were postulated to be an excellent biomarker of past exposures to relatively low doses of ionizing radiation that would make a welcome addition to the other cytogenetic rearrangements that were studied, i.e. such as translocations and dicentrics. It was hypothesized that inversions would be particularly indicative of previous exposure to Cosmic Galactic Radiation encountered by deep space voyagers. KromaTiD Inc. completed a convincing prototype of the methodology, i.e. a fluorescent chromatid paint to human chromosome three, and then demonstrated the predicted capability to detect inversions after exposure to ionizing radiation. Inversions appeared as fluorescent signal switches that appeared on the opposite chromatid of chromosome 3, opposite to their original position, allowing relative ease of scoring and suggesting to us that scoring might be automated completely in the future. We then used the assay to count the increased frequency of inversions that were produced after gamma radiation exposure, a proof of concept that the technology could be used for biodosimetry. Recently, as the result of a direct collaboration with NASA space radiation investigators, we described the production of chromosome inversions caused by high LET, HZE particles. Our results showed a dramatic dose-dependent increase in the yields of inversions that were induced by high LET charged particles, similar to those encountered in deep space as compared to low LET gamma rays. This omnibus award allowed us to develop new dGH chromatid paints to human chromosomes 1 and 2, and to recreate the dGH chromosome 3 chromatid paint using a new cheaper and faster methodology. The superior KromaTiD paints reported here have directly improved the performance of KromaTiD's biodosimetry assays while at the same time enabled a significant reduction in the cost per assay. This allows the rapid development of increasingly sensitive biodosimetry inversion assays to be used in the analysis of radiation risk for astronauts.
Bibliography Type:	Description: (Last Updated: 01/11/2021)
Articles in Peer-reviewed Journals	Ray FA, Robinson E, McKenna M, Hada M, George K, Cucinotta F, Goodwin EH, Bedford JS, Bailey SM, Cornforth MN. "Directional genomic hybridization: inversions as a potential biodosimeter for retrospective radiation exposure." Radiat Environ Biophys. 2014 May;53(2):255-63. Epub 2014 Jan 30. http://dx.doi.org/10.1007/s00411-014-0513-1 ; PubMed PMID: 24477407 , May-2014
Articles in Peer-reviewed Journals	Robinson E, McKenna MJ, Bedford JS, Goodwin EH, Cornforth MN, Bailey SM, Ray FA. "Directional genomic hybridization (dGH) for detection of intrachromosomal rearrangements." Methods Mol Biol. 2019;1984:107-16. https://doi.org/10.1007/978-1-4939-9432-8_13 ; PMID: 31267426 , Jul-2019